Nicolas “Mikayla” Fisher

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EDUCATION

Worcester Polytechnic Institute — Expected May 2030
B.S. Aerospace Engineering — Worcester, MA
Varsity Field Hockey, High-Powered Rocketry, AIAA, SWE

Monterey High School — 2022-2026
IB Diploma · Ranked 16 of 350 · 4.6 weighted GPA · IB HL: Math AA, Physics, Chemistry, English A
Varsity Field Hockey (Captain), FIRST Robotics, Monterey Academy of Oceanographic Science, Summa Cum Laude

TECHNICAL SKILLS

Engineering: Control systems (PID, feedforward, state estimation), FEA, MATLAB

CAD & Simulation: Onshape, SOLIDWORKS, Fusion 360 (CAM)

Manufacturing: CNC mill and router, manual lathe, FDM printing, machining, mechanical assembly

Software: Java, C, C++, Python, CUDA, ARM Assembly, Git, code review, CI/CD

EXPERIENCE

Johnson & Johnson MedTech — Summer 2026
Mechanical Engineering Intern — Santa Clara, CA
Mechanical Engineering Intern on the Haptic Interface Device (HID) team working R&D on the OTTAVA Surgical System.

- **Friction characterization fixture**
  - Developed an electromechanical test fixture for characterizing friction and hysteresis in HID subassemblies
  - Engineered a low-cost custom data acquisition solution using OTS microcontrollers and ADCs after pivoting from unavailable COTS DAQ hardware, achieving equivalent measurement capability
  - Implemented signal processing in analysis software, including Butterworth filtering, ensemble averaging with statistical outlier rejection, and FFT-based frequency-domain analysis of the friction signal
  - Applied statistical methods (t-tests, Signal to Noise ratio, confidence intervals) to validate measurement integrity and quantify system uncertainty
  - Created documentation of both the codebase and testing setup for verification and R&D testing
- Designed 10 manufacturing fixtures to aid with repeatability, alignment, and assembly speed of HID subassemblies.

WPI High Powered Rocketry Club — 2026-present
GNC Experimental Hardware & GNC Software — Worcester, MA

- **Robust Roll Control Under Canard-Induced Roll Reversal via Per-Canard Extremum Seeking** (Research paper)
  - Lead-researcher on a project studying roll-reversal effects on canard-controlled rockets for the team’s experimental rocket launch, developing a control strategy that’s robust to reversal through online measurement
  - Developing a novel, LQR-based control algorithm based on combining lessons from cross-disciplinary literature review, including using extremum seeking control to optimize control authority and using lessons from adaptive INDI (Smeur et. al), to combat reversal when it does occur.

FRC Team 4255 — Robodores — 2022 – 2026
President and Director of Engineering · FIRST Dean’s List Semi-Finalist (2×) — Monterey, CA

- Assumed full organizational leadership after the founding mentor's passing, rebuilding and stabilizing the program in one season with no adult oversight; grew membership 10→30 and taught 60+ students software, CAD, and manufacturing
- Recruited 5+ mentors from zero, including an Apple engineer whose partnership brought $100K in support, including an FRC practice field open to all Monterey Bay teams plus advanced manufacturing equipment, and launched an annual Week Zero scrimmage now hosting 12+ regional teams
- Built an in-house global pose localization stack from AprilTag detection, enabling shoot-from-anywhere scoring; extended it to a GTSAM factor-graph estimator with CUDA-accelerated detection alongside partner teams, then to shoot-on-the-move
- Self-taught control theory to implement motion-profiled PID with feedforward and a finite-state supervisor for a 2-DOF arm
- Sole mechanical designer of the 2025 robot after the design students graduated: 4-DOF, zero mechanism failures across the season, first regional win; 2026 design driven by FEA
- Implemented shoot-on-the-move in 2026 robot: forward-predicted robot pose across latency (shooter spin-up, response time), then used Euler’s method for a lead-angle aim point accounting for chassis velocity and projectile time-of-flight

LEADERSHIP AND VOLUNTEERING

- **Founder, Interim President, Secretary – Robodores Foundation (2024–Present):** Founded and incorporated a 501(c)(3) nonprofit organization to support STEM education in Monterey county. Responsible for managing board meetings and documentation.